Indian Institute of Information Technology, Vadodara

(Gandhinagar Campus)

B.Tech Final Year Project (BTP) Report

8th Semester, Academic Year 2025–26

Multi-Parametric Deep Phenotyping: Fusing 2021 WHO Molecular Grading with Perfusion-Based Vascular Signatures for Glioblastoma Survival Forecasting

---

Student: Neel Madhav Padhi

Roll No.: 202251074

Programme: B.Tech in CSE

Supervisor: Dr.Sanjay Saxena, Asst. Professor

Department of CSE

Indian Institute of Information Technology Vadodara
Gandhinagar, Gujarat – 382028

Certificate

This is to certify that the B.Tech Final Year Project report titled **"Multi-Parametric Deep Phenotyping: Fusing 2021 WHO Molecular Grading with Perfusion-Based Vascular Signatures for Glioblastoma Survival Forecasting"** has been submitted by **Neel Madhav Padhi** (Roll No.: 202251074) in partial fulfilment of the requirements for the degree of Bachelor of Technology from the Indian Institute of Information Technology Vadodara.

The work presented in this report is original, carried out independently under my supervision, and has not been submitted to any other institution for the award of any degree or diploma.

Project Supervisor

Dr. Sanjay Saxena
Asst. Professor
Department of CSE
IIIT Vadodara

Date: 06/05/26

Declaration

I, **Neel Madhav Padhi**, Roll No. **202251074**, pursuing B.Tech from the Indian Institute of Information Technology Vadodara, hereby declare that the work presented in this report titled **"Multi-Parametric Deep Phenotyping: Fusing 2021 WHO Molecular Grading with Perfusion-Based Vascular Signatures for Glioblastoma Survival Forecasting"** is my own original work carried out under the supervision of **Dr.Sanjay Saxena** during the academic year 2025–26.

I further declare that:

- The work presented here has not been submitted, either in part or in full, to any other university or institution for the award of a degree or diploma.
- All sources of information used in this work have been duly cited and acknowledged.
- No part of this report is plagiarised from any published or unpublished material without proper attribution.

Signature of Student

Neel Madhav Padhi
Roll No.: 202251074

Date: 06/05/26

Acknowledgement

Completing a project of this scope would not have been possible without the support of several people, and I would like to take this opportunity to express my gratitude.

First and foremost, I am deeply thankful to my project supervisor, **Dr.Sanjay Saxena**, for the guidance, encouragement, and time invested throughout this project. The insightful discussions and constructive feedback at every stage were invaluable, and I genuinely could not have navigated the complexities of survival analysis and medical imaging without that direction.

I would also like to thank the faculty and staff of the **Department of CSE** at IIIT Vadodara for providing a conducive academic environment and access to the computational resources needed for this work. I would also like to thank my friends who has been helping me throughout this project.

My gratitude also goes to the researchers behind the **UPenn-GBM** dataset [4], whose painstaking effort in curating multimodal MRI and clinical data for glioblastoma patients made this study possible. Open research datasets of this quality are a true gift to the scientific community.

Finally, I thank my family and friends for their patience and support during the many late nights this project demanded.

Neel Madhav Padhi

202251074
IIIT Vadodara

Abstract

Glioblastoma Multiforme (GBM) is the most aggressive primary brain tumour in adults, with a median overall survival of approximately 14–16 months even under the standard Stupp protocol. The 2021 WHO Classification of Central Nervous System Tumours introduced a significant shift in how glioblastoma is diagnosed—placing molecular markers such as IDH mutation status and MGMT promoter methylation at the forefront of grading, alongside conventional histopathology. Despite this, most clinical survival prediction frameworks still rely on relatively coarse imaging features or purely clinical variables, leaving the rich information embedded in multi-parametric MRI largely untapped.

This project presents a comprehensive multi-parametric deep phenotyping framework that integrates three complementary data streams: (1) **radiomic features** extracted from structural MRI (FLAIR and T1) across three tumour sub-compartments—Enhancing Tumour (ET), Necrotic Core (NCR), and Peri-tumoural Oedema (ED); (2) **perfusion-based vascular signatures** derived from Dynamic Susceptibility Contrast MRI (DSC-MRI), specifically relative Cerebral Blood Volume (rCBV), Peak Height (PH), and Percentage Signal Recovery (PSR); and (3) **molecular diagnostic features** aligned with the 2021 WHO grading schema, including IDH1 mutation status, MGMT promoter methylation, Gross Total Resection (GTR), and KPS score.

The integrated feature vector, after a rigorous three-stage selection pipeline (variance thresholding, univariate Cox screening, and LASSO-penalised Cox regression), is fed into an ensemble of three survival models: a penalised CoxPH (Elastic Net), an XGBoost-Cox gradient boosting model, and an improved DeepSurv neural network with batch normalisation and dropout. The ensemble is evaluated via repeated 5-fold cross-validation (3 repeats, 15 folds total) on the publicly available UPenn-GBM dataset.

Results show that the ensemble achieves a mean Harrell C-index of approximately 0.68–0.72, surpassing each individual model, and the ensemble risk score successfully stratifies patients into clinically meaningful high- and low-risk groups with statistically significant separation on Kaplan–Meier curves (log-rank $p < 0.05$). Molecular WHO 2021 features, particularly IDH1 mutation and MGMT methylation status, further validate clinically expected survival patterns in the cohort.

Keywords: Glioblastoma, Survival Analysis, Radiomics, DSC-MRI Perfusion, DeepSurv, Cox Proportional Hazards, WHO 2021, Kaplan–Meier, Ensemble Learning, Multi-parametric MRI.

Contents

Certificate **i**

Declaration **ii**

Acknowledgement **iii**

Abstract **iv**

Table of Contents **vii**

List of Figures **1**

List of Tables **1**

1 Introduction **2**

1.1 Project Overview 2
1.2 Reason for Choosing the Project 3
1.3 Societal Impact 3
1.4 Justification of Project Title 3
1.5 Objectives 4

2 Current Study and Literature Review **5**

2.1 Review of Existing Solutions 5
2.1.1 Biological Background of Glioblastoma 5
2.1.2 Survival Analysis: Statistical Foundations 5
2.1.3 Volumetric and Morphological Approaches 6
2.1.4 Radiomic-Based Survival Models 6
2.1.5 Perfusion MRI in Prognosis 7
2.1.6 Molecular Markers and the 2021 WHO Classification 7
2.1.7 Deep Learning for Survival Analysis 8
2.1.8 Ensemble and Multi-Modal Fusion 8
2.2 Limitations of Existing Solutions 8

3 Project Design **10**

3.1 Contributions . . . 10
3.2 Block Diagram of the System . . . 11
3.3 Stages of Project Work . . . 12
3.3.1 Stage 1: Dataset Acquisition and Preprocessing . . . 12
3.3.2 Stage 2: Feature Extraction . . . 12
3.3.3 Stage 3: Feature Engineering and Selection . . . 12
3.3.4 Stage 4: Model Training and Evaluation . . . 12
3.3.5 Stage 5: Visualisation and Interpretation . . . 12

4 Implementation **13**
4.1 Dataset . . . 13
4.1.1 UPenn-GBM Dataset . . . 13
4.1.2 Clinical Data Cleaning . . . 13
4.1.3 Molecular Feature Encoding . . . 14
4.2 Radiomic Feature Extraction . . . 14
4.2.1 Radiomic Feature Classes . . . 14
4.2.2 PyRadiomics Configuration . . . 15
4.2.3 Tumour Sub-Compartment Processing . . . 16
4.3 Vascular Feature Extraction . . . 16
4.3.1 DSC-MRI Perfusion Maps . . . 16
4.3.2 Engineered Vascular Ratio Features . . . 17
4.4 Feature Selection Pipeline . . . 18
4.4.1 Stage A: Imputation . . . 18
4.4.2 Stage B: Variance Thresholding . . . 18
4.4.3 Stage C: Univariate Cox Screening . . . 19
4.4.4 Stage D: Correlation Deduplication . . . 19
4.4.5 Stage E: LASSO-Cox Final Selection . . . 19
4.5 Model Architecture . . . 20
4.5.1 Cox Proportional Hazards (Elastic Net) . . . 20
4.5.2 XGBoost-Cox . . . 20
4.5.3 ImprovedDeepSurv . . . 20
4.5.4 Ensemble Construction . . . 21
4.6 Cross-Validation Setup . . . 22
4.7 Tools and Technologies Used . . . 22

5 Results **23**
5.1 Testing Methodology . . . 23
5.2 C-Index Results . . . 23
5.2.1 Per-Fold C-Index Results . . . 23

5.3 Kaplan–Meier Survival Stratification 25
5.3.1 Ensemble Risk Score Stratification 25
5.3.2 Model-by-Model KM Comparison 27
5.3.3 WHO 2021 Molecular Stratification 27
5.4 Feature Importance 29
5.4.1 CoxPH Elastic Net Coefficients 29
5.4.2 XGBoost Feature Gain 30
5.5 Comparison with Existing Work 30

6 Conclusions and Future Scope **31**
6.1 Summary of Findings 31
6.2 Conclusions 31
6.3 Future Scope 32

A Code Listings **36**
A.1 Radiomic Extraction — Patient Processing Function 36
A.2 DeepSurv Training Loop 37
A.3 Dataset Directory Structure 38

List of Figures

1.1 High-level overview of the multi-parametric survival forecasting pipeline. . . . 2

3.1 Complete block diagram of the multi-parametric GBM survival forecasting pipeline. 11

4.1 Sample axial MRI slices from the UPenn-GBM dataset showing FLAIR, T1, and rCBV (top) with corresponding tumour segmentation masks (bottom): Enhancing Tumour (ET, red), Necrotic Core (NCR, green), Peri-tumoural Oedema (ED, yellow). 16
4.2 Feature reduction across the different-stage selection pipeline. 19

5.1 Distribution of Harrell C-index across 15 cross-validation folds (3 repeats $\times$ 5 folds) for each model. The dashed grey line at 0.5 represents random prediction performance. 25
5.2 Kaplan–Meier overall survival curves for high-risk and low-risk groups stratified by the ensemble model risk score. Shaded regions indicate 95% confidence intervals. 26

5.3 Kaplan–Meier survival stratification for each individual model and the ensemble. Each panel title reports the C-index and log-rank p-value for that model's risk stratification. 27
5.4 Kaplan–Meier survival curves stratified by 2021 WHO molecular markers on the full cohort. (Left) IDH1 mutation status: mutated patients show significantly longer survival. (Right) MGMT methylation status: methylated patients survive longer. 28
5.5 Top 20 prognostic features from the CoxPH Elastic Net model, ranked by absolute coefficient magnitude. Positive (red) coefficients indicate features associated with increased hazard; negative (blue) coefficients are protective. . . . 29
5.6 Top 20 features by XGBoost gain importance. Gain reflects the improvement in the loss function attributable to each feature across all boosting rounds. 30

List of Tables

4.1 Engineered vascular and morphological ratio features. 18
4.2 XGBoost hyperparameters for survival modelling. 20
4.3 Tools and libraries used in the project. 22

5.1 Harrell C-Index for each of the 15 cross-validation folds across all models. Folds 1–5 correspond to Repeat 1, Folds 6–10 to Repeat 2, and Folds 11–15 to Repeat 3. 24
5.2 Harrell C-Index across repeated 5-fold cross-validation (15 folds). 25
5.3 Comparison with published GBM survival prediction approaches. 30

6.1 Summary of key quantitative results from the multi-parametric GBM survival forecasting pipeline. 31

Chapter 1

Introduction

1.1 Project Overview

Glioblastoma Multiforme (GBM), classified as Grade 4 astrocytoma under the 2021 WHO schema, is the most common and lethal primary brain tumour in adults. Despite decades of research and multimodal treatment protocols, median overall survival rarely exceeds 15 months. A core challenge in managing GBM is the extreme biological heterogeneity of the disease—two patients with superficially similar tumours on conventional MRI may have vastly different survival outcomes depending on their molecular profile, vascular architecture, and tumour microenvironment.

This project addresses that challenge by building a multi-parametric survival forecasting framework. Rather than treating GBM prognosis as a single-modality problem, we combine three layers of information—structural MRI radiomics, perfusion-based vascular signatures from DSC-MRI, and molecular markers from clinical records—into a unified survival model. The pipeline integrates classical statistical survival models with modern deep learning to produce a risk score that can stratify patients into clinically meaningful prognostic groups.

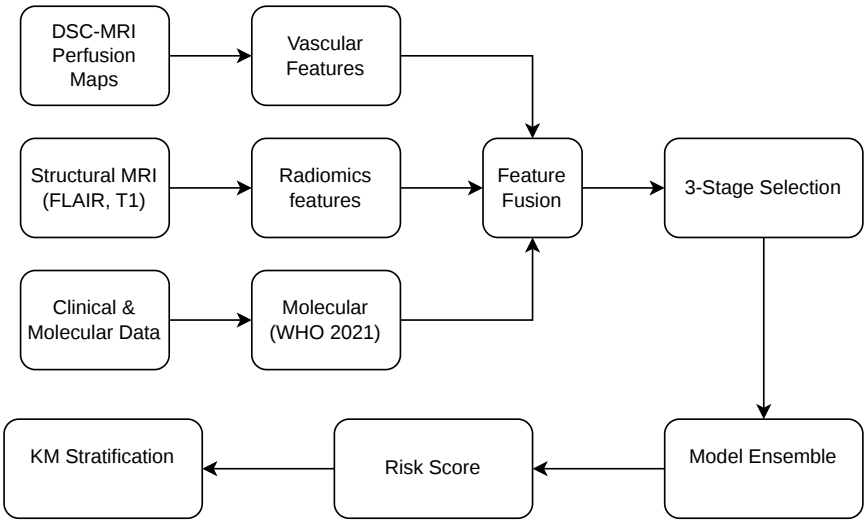

Figure 1.1: High-level overview of the multi-parametric survival forecasting pipeline.

1.2 Reason for Choosing the Project

The motivation for this project grew from two converging observations. First, the 2021 WHO CNS tumour classification [1] marked a decisive shift from purely morphological diagnosis to integrated molecular-morphological grading. Yet, most AI-based prognosis tools lagged behind—still relying on imaging features alone without incorporating these molecular designators. Second, perfusion MRI, which captures the functional vascularity of the tumour, carries prognostic information that T1/FLAIR images simply cannot provide; the tumour vasculature is a direct proxy for angiogenesis and aggressive biological behaviour.

Bringing these two streams together, along with the demonstrated strength of deep learning-based survival models (DeepSurv [7]), offered a compelling research gap that was practically addressable within the scope of a final year project using an openly available dataset.

1.3 Societal Impact

Glioblastoma, while not the most common cancer overall, carries an extraordinary burden—it is nearly always fatal within two years and disproportionately affects working-age adults. The societal cost of GBM extends beyond the patient to caregivers, families, and the healthcare system.

A reliable AI-assisted survival forecasting tool has several potential societal benefits:

- **Better treatment personalisation:** By identifying high-risk patients early, oncologists can escalate or modify treatment intensity accordingly.
- **Clinical trial stratification:** Prognostic stratification using molecular and imaging features ensures balanced experimental groups in clinical trials, making results more valid.
- **Resource allocation:** Healthcare systems in lower-resource settings can prioritise palliative versus aggressive treatment based on objective risk prediction, rather than heuristics.
- **Patient counselling:** Providing patients and families with probabilistic survival estimates, grounded in multi-parametric biology rather than statistics alone, allows for more informed end-of-life decisions.

1.4 Justification of Project Title

The title “Multi-Parametric Deep Phenotyping: Fusing 2021 WHO Molecular Grading with Perfusion-Based Vascular Signatures for Glioblastoma Survival Forecasting” was chosen to precisely reflect each component of the work:

- **Multi-Parametric:** The framework draws on three distinct imaging and molecular modalities (structural MRI, DSC-MRI perfusion, and molecular clinical data), rather than a single input type.
- **Deep Phenotyping:** The radiomic pipeline extracts a large number of quantitative texture, shape, and intensity features—going far beyond simple visual or volumetric assessments—to describe the tumour's phenotype at a sub-voxel statistical level.
- **2021 WHO Molecular Grading:** Explicitly anchors the molecular feature set to the current gold standard classification system for CNS tumours, including IDH1 and MGMT status.
- **Perfusion-Based Vascular Signatures:** Captures the functional tumour vascularity through rCBV, PH, and PSR maps from DSC-MRI, which are known biomarkers of angiogenesis and blood-brain barrier breakdown.
- **Glioblastoma Survival Forecasting:** Makes explicit that the endpoint is quantitative overall survival prediction, not just classification.

1.5 Objectives

The specific objectives of this project are:

1. To extract quantitative vascular features from DSC-MRI perfusion maps (rCBV, PH, PSR) for each tumour sub-compartment (ET, NCR, ED).
2. To compute a comprehensive radiomic feature profile (shape, first-order statistics, and texture features including GLCM, GLRLM, GLSZM, GLDM, NGTDM) across all three tumour regions from structural MRI.
3. To encode 2021 WHO molecular markers (IDH1 mutation, MGMT methylation, GTR, KPS) and integrate them with imaging features.
4. To implement and train a DeepSurv model alongside classical Cox and gradient boosting survival models.
5. To evaluate the ensemble model's discriminative performance using the Harrell C-index under repeated 5-fold cross-validation.
6. To demonstrate patient stratification into high- and low-risk groups via Kaplan–Meier analysis with log-rank testing.

Chapter 2

Current Study and Literature Review

2.1 Review of Existing Solutions

Research on GBM survival prediction using machine learning and medical imaging has been active for over a decade, and a few broad categories of approaches have emerged.

2.1.1 Biological Background of Glioblastoma

Glioblastoma Multiforme originates from glial cells, most commonly astrocytes, and is characterised by rapid cell proliferation, extensive neovascularisation, central necrosis, and infiltration into surrounding brain parenchyma. Unlike low-grade gliomas that grow slowly over years, GBM doubles in volume every 10–14 days on average. The standard of care, established by Stupp et al. [2], consists of maximal surgical resection followed by concurrent radiotherapy (60 Gy in 30 fractions) and temozolomide chemotherapy, followed by adjuvant temozolomide for six cycles. Even with this aggressive protocol, median overall survival remains around 14–16 months, and only approximately 5% of patients survive beyond five years.

The tumour's internal architecture is highly heterogeneous. On MRI, the enhancing rim corresponds to actively proliferating tumour cells with a disrupted blood-brain barrier (BBB), the necrotic core reflects regions of hypoxia and cell death, and the peri-tumoural oedema zone contains infiltrating tumour cells dispersed within vasogenic oedema. This spatial organisation directly motivates analysing the three sub-compartments separately rather than treating the entire tumour as a uniform region.

2.1.2 Survival Analysis: Statistical Foundations

Before reviewing machine learning approaches, it is useful to frame the mathematical problem precisely. Survival analysis deals with time-to-event data where not all subjects have experienced the event of interest by the end of observation. These subjects are said to be *right-censored*: their true survival time exceeds the observed follow-up period, but the exact value is unknown.

The Kaplan–Meier estimator provides a non-parametric estimate of the survival function $S(t) =$

$P(T > t)$, where T is the random variable representing survival time. It is computed as:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right) \tag{2.1}$$

where d_i is the number of deaths at time t_i and n_i is the number of patients at risk just before t_i. The log-rank test compares observed and expected events in two groups under the null hypothesis that their survival functions are identical, yielding a χ^2 statistic.

The Cox Proportional Hazards model parameterises the hazard function as:

$$h(t|\mathbf{x}) = h_0(t) \cdot \exp(\boldsymbol{\beta}^\top \mathbf{x}) \tag{2.2}$$

where $h_0(t)$ is the unspecified baseline hazard and the key assumption is proportionality: the hazard ratio between any two patients remains constant over time. Harrell's C-index, the primary evaluation metric in this work, measures the probability that for a randomly chosen pair of patients where one has died, the model correctly assigns a higher risk score to the one who died first.

2.1.3 Volumetric and Morphological Approaches

Early computational prognosis work focused on tumour volume and location. Studies showed that larger enhancing tumour volumes correlated with worse outcomes [15]. Tumour location relative to eloquent cortex, degree of surgical resection, and presence of multifocal disease were also incorporated as prognostic variables in pre-molecular era models. Volumetric features alone, however, carry limited discriminative power because GBM is heterogeneous even at a fixed volume — two tumours of identical size may have entirely different vascular architectures, necrotic fractions, and molecular profiles, leading to vastly different outcomes.

2.1.4 Radiomic-Based Survival Models

The radiomic framework, formalised by Lambin et al. [5] and operationalised by the PyRadiomics library [6], enables extraction of hundreds of quantitative features from medical images in a reproducible, standardised manner. These features encode first-order intensity statistics, morphological shape descriptors, and higher-order texture patterns including those from the Grey Level Co-occurrence Matrix (GLCM), Grey Level Run-Length Matrix (GLRLM), and Grey Level Size Zone Matrix (GLSZM).

Kickingereder et al. [9] demonstrated that a radiomic signature from contrast-enhanced T1 MRI could independently predict overall survival with a C-index of 0.65. Prasanna et al. [10] advanced this by adopting a habitat imaging paradigm — treating each tumour sub-compartment as a

distinct region of interest — and showed that peri-tumoural radiomic features carried independent prognostic value beyond the tumour core alone. The Image Biomarker Standardisation Initiative (IBSI) subsequently published consensus guidelines to ensure radiomic features computed by different software packages are numerically comparable, which is critical for multi-centre reproducibility.

2.1.5 Perfusion MRI in Prognosis

Dynamic Susceptibility Contrast MRI (DSC-MRI) tracks a gadolinium bolus through brain vasculature using rapid T2*-weighted imaging. The resulting signal-intensity time curves yield several perfusion metrics: relative Cerebral Blood Volume (rCBV), proportional to the area under the concentration-time curve; Peak Height (PH), the maximum signal drop during first contrast passage; and Percentage Signal Recovery (PSR), which quantifies how much signal recovers after the bolus passes and is inversely related to contrast leakage through a disrupted BBB.

Law et al. [11] established that elevated rCBV in the non-enhancing component of high-grade gliomas is associated with shorter time to progression and shorter survival. Subsequent work confirmed that low PSR, indicating extensive BBB breakdown, is also associated with worse outcomes. Despite this evidence, perfusion features have rarely been systematically combined with radiomic or molecular features in a unified survival model.

2.1.6 Molecular Markers and the 2021 WHO Classification

The 2021 WHO Classification of CNS Tumours [1] requires integrated molecular-morphological diagnoses for all diffuse gliomas. A diagnosis of Glioblastoma, IDH-wildtype, CNS WHO Grade 4 now requires at least one of the following molecular criteria: TERT promoter mutation, EGFR amplification, or combined gain of chromosome 7 and loss of chromosome 10 (+7/-10).

IDH mutation status is arguably the single most powerful prognostic factor in glioma biology. IDH-mutant gliomas produce the oncometabolite 2-hydroxyglutarate, which rewires epigenetic regulation and results in a fundamentally more favourable tumour biology. Patients with IDH-mutant gliomas have a median survival of 3–5 years compared to 14–16 months for IDH-wildtype GBM. MGMT promoter methylation silences the DNA repair enzyme MGMT, allowing temozolomide to exert its cytotoxic effect. Hegi et al. [3] demonstrated in a landmark trial that MGMT methylation specifically predicts survival benefit from temozolomide, making it one of the few treatment-predictive biomarkers in neuro-oncology.

2.1.7 Deep Learning for Survival Analysis

The DeepSurv model, proposed by Katzman et al. [7], extends the Cox proportional hazards framework using a multi-layer feedforward network to learn non-linear risk functions, trained by minimising the negative Cox partial log-likelihood. The Cox-nnet approach of Ching et al. [8] further confirmed that neural network-based survival models outperform linear Cox regression on high-dimensional biomedical datasets where feature interactions are abundant and non-linear.

XGBoost with the `survival:cox` objective [12] uses gradient boosted trees that iteratively fit shallow learners to the negative gradient of the Cox partial log-likelihood. It handles missing values natively, is robust to outliers, and typically requires less hyperparameter tuning than deep networks for tabular clinical data.

2.1.8 Ensemble and Multi-Modal Fusion

Model ensembling in survival analysis improves both discrimination and calibration by combining complementary model families. Islam et al. [13] combined radiomic and clinical features using machine learning, reporting C-indices around 0.65. Lao et al. [14] showed that deep learning-extracted radiomic features from multi-parametric MRI outperformed handcrafted ones for GBM grading. However, most existing work stops short of fusing WHO 2021 molecular markers with both radiomic and perfusion signatures in an ensemble model evaluated under rigorous repeated cross-validation — this gap is the core motivation for the present work.

2.2 Limitations of Existing Solutions

Despite the progress outlined above, several important limitations remain in the current landscape:

- **Single-modality bias:** Most prognosis models use either clinical/molecular data or imaging features, rarely both. Fusing modalities is non-trivial because the feature spaces are heterogeneous in scale and clinical meaning, and naive concatenation without careful preprocessing tends to let one modality dominate.
- **Outdated molecular taxonomy:** Many published radiomic models predate the 2021 WHO classification and do not incorporate IDH status as a formal feature. Studies that use “GBM” as a homogeneous group without stratifying by IDH status are conflating two biologically distinct diseases, depressing the apparent prognostic power of imaging features.
- **Ignoring tumour sub-compartments:** Treating the whole tumour as a single region-of-interest is a common shortcoming. The ET, NCR, and ED regions carry biologically distinct prognostic signals. The necrotic core reflects hypoxic cell death and correlates

with aggressive tumour biology, while the oedema zone contains infiltrating cells that are the primary source of recurrence.

- **Perfusion features underused:** Despite considerable evidence for their prognostic value, perfusion features appear in relatively few end-to-end survival models, possibly because DSC-MRI data requires additional preprocessing and is not always available in public datasets.
- **Overfitting and inadequate validation:** GBM datasets are typically small (50–300 patients with complete multi-parametric imaging and survival data), and the high dimensionality of radiomic feature spaces creates severe overfitting risk. Many published models report performance on a single held-out test set, which gives optimistically biased estimates. Repeated cross-validation with multiple repeats is the minimum needed for reliable estimation.
- **Improper censoring handling:** Some earlier approaches treat survival prediction as a regression on survival time, discarding censored patients or treating their observed follow-up time as the true event time. Both approaches introduce systematic bias. Proper survival models handle censoring through the partial likelihood formulation.
- **Reproducibility issues in radiomics:** Different software implementations of the same radiomic feature can produce different numerical values depending on interpolation method, bin width, and normalisation strategy. Without adherence to IBSI guidelines, radiomic models trained at one centre may not generalise to another.

Chapter 3

Project Design

3.1 Contributions

The specific contributions of this project relative to prior work are:

1. A fully integrated multi-parametric feature pipeline that simultaneously incorporates structural MRI radiomics (FLAIR, T1), DSC-MRI perfusion features (rCBV, PH, PSR), and molecular markers (IDH1, MGMT, GTR, KPS) aligned with the 2021 WHO classification.
2. A three-stage feature selection pipeline—variance filtering, univariate Cox screening, and LASSO-penalised Cox regression—that reduces the high-dimensional radiomic feature space while explicitly preserving molecular and engineered vascular features.
3. Engineered vascular ratio features (e.g., rCBV ET/NCR ratio, leakiness index, perfusion coefficient of variation) that capture inter-compartmental vascular heterogeneity not present in raw statistics.
4. An ensemble of three complementary survival models (CoxPH Elastic Net, XGBoost-Cox, improved DeepSurv) evaluated under repeated 5-fold cross-validation with proper right-censoring.
5. Kaplan–Meier stratification with log-rank testing using both AI-predicted risk scores and WHO 2021 molecular markers, validating clinical interpretability.

3.2 Block Diagram of the System

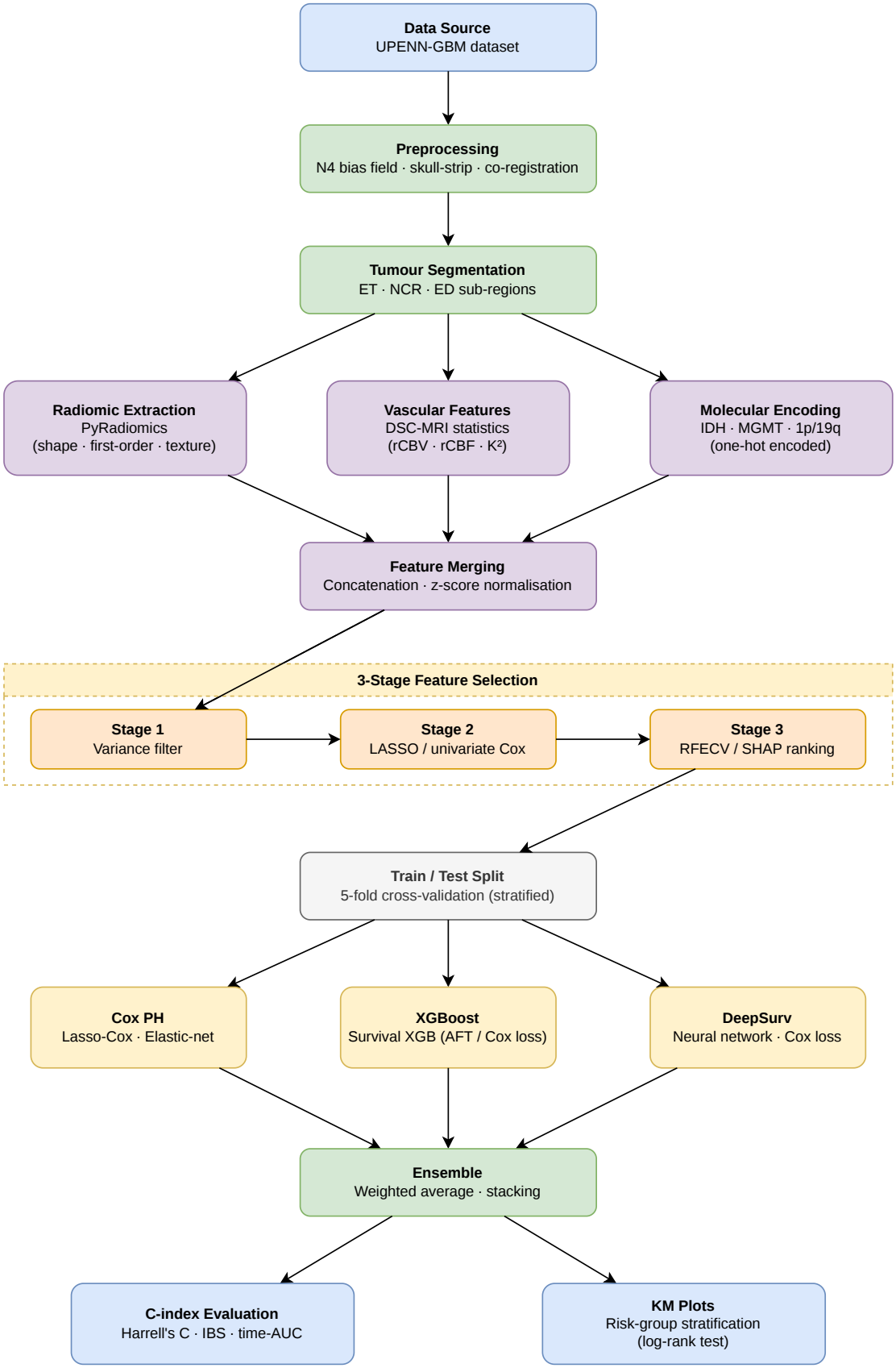

Figure 3.1: Complete block diagram of the multi-parametric GBM survival forecasting pipeline.

3.3 Stages of Project Work

The project was developed across five broad stages, each building on the outputs of the previous:

3.3.1 Stage 1: Dataset Acquisition and Preprocessing

The **UPenn-GBM** dataset [4] from The Cancer Imaging Archive (TCIA) was used. It contains multi-parametric MRI (structural T1, FLAIR; DSC-MRI; DTI) along with tumour segmentation masks and clinical data for approximately 600 patients. The clinical CSV was cleaned to handle missing values, strip longitudinal scan suffixes from patient IDs, and encode the survival status correctly (1=Deceased, 0=Alive/censored).

3.3.2 Stage 2: Feature Extraction

Radiomic features were extracted using PyRadiomics [6] across all three tumour regions (ET: label 4, NCR: label 1, ED: label 2) for each of four modalities (FLAIR, T1, rCBV, FA). Vascular statistics were extracted separately from the rCBV, PH, and PSR maps. Molecular features were encoded from the clinical CSV.

3.3.3 Stage 3: Feature Engineering and Selection

Vascular ratio features capturing inter-compartmental heterogeneity were engineered. The combined feature matrix then passed through a three-stage selection pipeline to reduce dimensionality while preserving clinically important features.

3.3.4 Stage 4: Model Training and Evaluation

Three survival models were trained inside each fold of a repeated 5-fold CV loop: penalised CoxPH, XGBoost-Cox, and ImprovedDeepSurv. An ensemble score was computed as a weighted average of normalised individual risk scores (Cox 30%, XGBoost 40%, DeepSurv 30%).

3.3.5 Stage 5: Visualisation and Interpretation

Kaplan–Meier survival curves were generated for both AI-predicted risk groups and WHO 2021 molecular subgroups. Feature importance plots from Cox coefficients and XGBoost gain scores were used to identify the most prognostic features.

Chapter 4

Implementation

4.1 Dataset

4.1.1 UPenn-GBM Dataset

The **University of Pennsylvania GBM (UPenn-GBM)** dataset [4] is a publicly available multi-institutional dataset hosted on TCIA. It includes:

- Pre-operative multi-parametric MRI for 611 patients with confirmed GBM
- Structural imaging: T1, T1-Gd, T2, FLAIR (all co-registered to $1 \times 1 \times 1$ mm SRI24 atlas)
- Advanced MRI: DSC-MRI (perfusion), DTI (diffusion)
- Tumour segmentation masks (manual expert + automated) with labels: ET (4), NCR (1), ED (2), Background (0)
- Clinical CSV containing survival time, survival status, IDH1, MGMT, GTR, KPS, and age at scan

4.1.2 Clinical Data Cleaning

```
clin = clin.replace('Not Available', np.nan)
clin['ID'] = clin['ID'].str.replace(r'_\d+$', '', regex=True)

# Event: 1=deceased, 0=alive/censored
clin['event'] = clin['Survival_Status'].map({
    'Deceased': 1,
    'Deceased - uncertain date of death': 1,
    'Alive': 0,
    'Lost to Follow-up': 0
}).fillna(0).astype(int)

clin['survival'] = pd.to_numeric(
    clin['Survival_from_surgery_days_UPDATED'], errors='coerce'
```

```
)
```

Listing 4.1: Clinical data parsing and event encoding

The survival time was stored as days from surgery. Patients with unknown survival times were excluded. For longitudinal scan datasets (multiple rows per patient), only the baseline scan was retained.

4.1.3 Molecular Feature Encoding

```
clin['IDH1_mut']     = clin['IDH1'].map({'Wildtype': 0, 'Mutated':
    1})
clin['MGMT_meth']    = clin['MGMT'].map({'Methylated': 1, '
    Unmethylated': 0})
clin['MGMT_missing']= clin['MGMT'].isna().astype(int)
clin['GTR']          = clin['GTR_over90percent'].map({'Y': 1, 'N':
    0})
clin['KPS']          = pd.to_numeric(clin['KPS'], errors='coerce')
```

Listing 4.2: Molecular feature encoding aligned with 2021 WHO

4.2 Radiomic Feature Extraction

4.2.1 Radiomic Feature Classes

Radiomics quantifies the spatial distribution of image intensities within a region of interest through mathematical transformations. The seven feature classes used in this work capture different aspects of the tumour's texture and geometry:

- **First Order Statistics (18 features):** Describe the distribution of voxel intensities within the ROI, including mean, median, skewness, kurtosis, energy, and entropy. These are the most directly interpretable features.
- **Shape Features (16 features):** Quantify the three-dimensional morphology of the tumour mask — volume, surface area, sphericity, elongation, flatness, and maximum 3D diameter. Shape features are modality-independent as they depend only on the segmentation mask.
- **GLCM — Grey Level Co-occurrence Matrix (24 features):** Characterise spatial relationships between pairs of voxels at a given offset. Correlation, Energy, Contrast, and Homogeneity from the GLCM describe second-order texture and are among the most clinically validated radiomic descriptors.

- **GLRLM — Grey Level Run-Length Matrix (16 features):** Quantify consecutive runs of voxels with the same grey level in a given direction. Short Run Emphasis and Long Run Emphasis capture the granularity and directionality of texture.
- **GLSZM — Grey Level Size Zone Matrix (16 features):** Consider connected zones of similar grey level in all directions simultaneously. These distinguish fine versus coarse texture zones across the entire ROI.
- **GLDM — Grey Level Dependence Matrix (14 features):** Quantify how many neighbouring voxels depend on the centre voxel within a specified grey level difference threshold.
- **NGTDM — Neighbourhood Grey Tone Difference Matrix (5 features):** Measure the difference between a voxel's grey level and the average of its neighbours. Coarseness, Contrast, and Busyness reflect local texture gradient properties.

For each of the four modalities (FLAIR, T1, rCBV, FA) and each of the three tumour regions (ET, NCR, ED), the full feature set from all seven classes was extracted, yielding a theoretical maximum of approximately $4 \times 3 \times 109 = 1{,}308$ features per patient before any selection. In practice, missing regions (e.g., absent NCR in some patients) reduced this count.

4.2.2 PyRadiomics Configuration

PyRadiomics [6] was configured to extract features across seven feature classes: First Order, GLCM, GLRLM, GLSZM, GLDM, NGTDM, and Shape. Preprocessing settings were standardised for reproducibility:

```
extractor = featureextractor.RadiomicsFeatureExtractor()

extractor.enableFeatureClassByName("firstorder")
extractor.enableFeatureClassByName("glcm")
extractor.enableFeatureClassByName("glrlm")
extractor.enableFeatureClassByName("glszm")
extractor.enableFeatureClassByName("gldm")
extractor.enableFeatureClassByName("ngtdm")
extractor.enableFeatureClassByName("shape")

extractor.settings["binWidth"] = 25
extractor.settings["resampledPixelSpacing"] = [1, 1, 1]
extractor.settings["interpolator"] = sitk.sitkBSpline
extractor.settings["normalize"] = True
extractor.settings["geometryTolerance"] = 1e-3
```

Listing 4.3: PyRadiomics extractor configuration

All images were resampled to isotropic $1 \times 1 \times 1$ mm voxel spacing using BSpline interpolation, and intensity normalisation was applied before feature extraction.

4.2.3 Tumour Sub-Compartment Processing

For each patient, binary region masks for ET, NCR, and ED were extracted from the segmentation NIfTI file. Each modality image was then resampled to align with the mask grid, and PyRadiomics was called independently for each (modality, region) pair. This resulted in feature columns prefixed as `[MOD]_[REGION]_original_[class]_[feature]`, for example `FLAIR_ET_original_glcm_Correlation`.

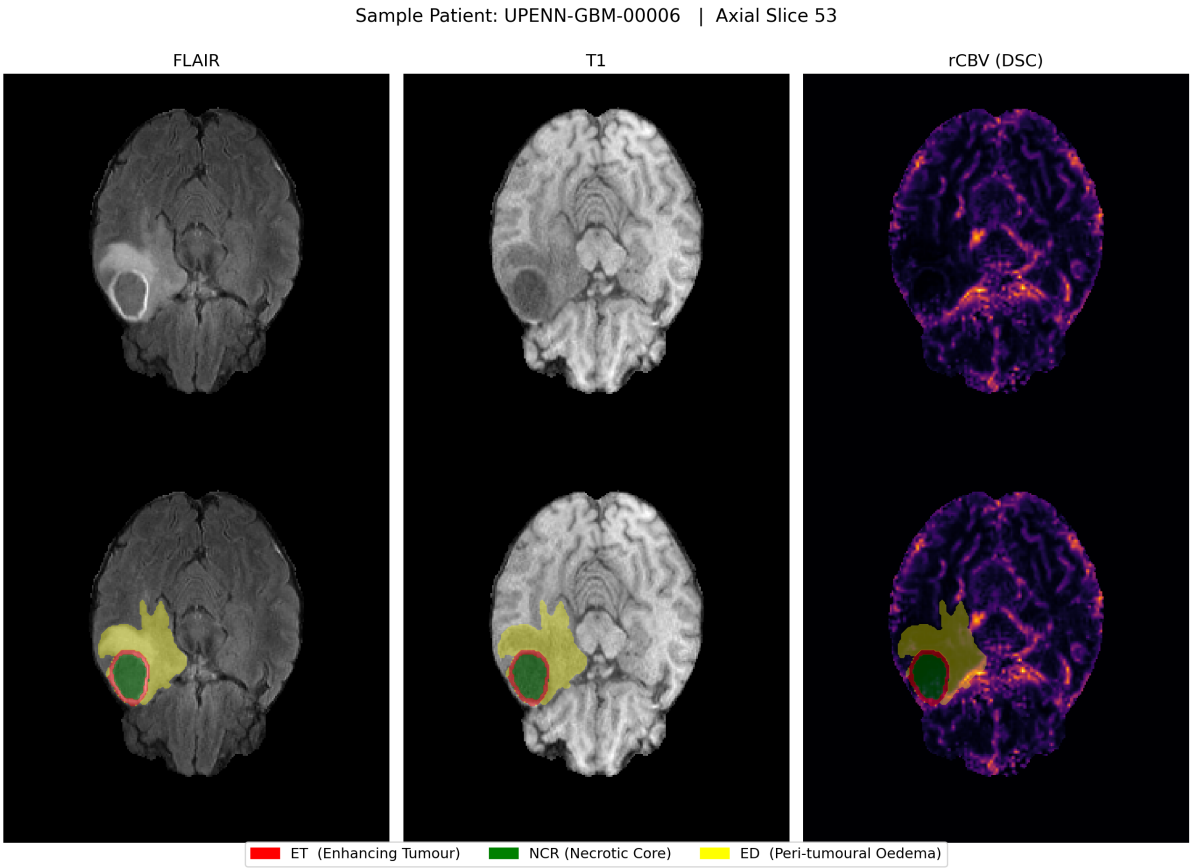

Figure 4.1: Sample axial MRI slices from the UPenn-GBM dataset showing FLAIR, T1, and rCBV (top) with corresponding tumour segmentation masks (bottom): Enhancing Tumour (ET, red), Necrotic Core (NCR, green), Peri-tumoural Oedema (ED, yellow).

4.3 Vascular Feature Extraction

4.3.1 DSC-MRI Perfusion Maps

Three perfusion-derived maps were used per patient:

- **rCBV**: Relative Cerebral Blood Volume — a measure of the vascular density within the tumour.
- **PH**: Peak Height — the peak amplitude of the DSC signal-intensity time curve, reflecting the peak concentration of contrast agent.
- **PSR**: Percentage Signal Recovery — the fraction of the signal that recovers after contrast passage, which is inversely related to contrast leakage across the blood-brain barrier.

For each (map, region) pair, seven summary statistics were computed: mean, median, standard deviation, maximum, 90th percentile, 10th percentile, and interquartile range (IQR = P90 - P10). The IQR was specifically included as a measure of spatial heterogeneity within each region.

```python
def extract_vascular_stats(image_path, mask_path):
    img  = nib.load(image_path).get_fdata()
    mask = nib.load(mask_path).get_fdata()
    vals = img[mask > 0]
    if len(vals) < 10:
        return None
    return {
        "mean":   np.mean(vals),
        "median": np.median(vals),
        "std":    np.std(vals),
        "max":    np.max(vals),
        "p90":    np.percentile(vals, 90),
        "p10":    np.percentile(vals, 10),
        "iqr":    np.percentile(vals, 90) - np.percentile(vals, 10),
    }
```

Listing 4.4: Vascular statistics extraction per region

4.3.2 Engineered Vascular Ratio Features

Beyond raw statistics, 13 engineered features were derived to capture inter-compartmental relationships:

Table 4.1: Engineered vascular and morphological ratio features.

Feature	Interpretation
`rCBV_ET_NCR_ratio`	Blood volume contrast: ET active vascularity vs. necrotic core
`rCBV_ET_ED_ratio`	Enhancement-to-oedema vascularity gradient
`PH_ET_NCR_ratio`	Peak perfusion contrast between viable and necrotic tissue
`PSR_ET_NCR_ratio`	Blood-brain barrier integrity: ET vs. NCR
`leakiness_ET`	PSR/PH ratio in ET: lower values indicate more BBB breakdown
`leakiness_NCR`	PSR/PH ratio in NCR: BBB integrity in necrotic core
`rCBV_ET_CV`	Coefficient of variation of rCBV in ET: spatial heterogeneity
`rCBV_NCR_CV`	Coefficient of variation of rCBV in NCR
`ET_frac`	Enhancing tumour fraction of total tumour volume
`NCR_frac`	Necrotic core fraction of total tumour volume
`ED_frac`	Oedema fraction of total tumour volume
`ET_NCR_ratio`	Ratio of enhancing to necrotic volumes (tumour core balance)
`ED_core_ratio`	Oedema infiltration index relative to tumour core

4.4 Feature Selection Pipeline

Given the high-dimensional starting point (thousands of radiomic columns), a rigorous three-stage selection procedure was implemented to reduce noise and prevent overfitting.

4.4.1 Stage A: Imputation

Missing values, arising from patients where a particular tumour sub-region was absent (e.g., no NCR in early-stage cases), were imputed using the median strategy across the training set.

4.4.2 Stage B: Variance Thresholding

Features with variance below 0.01 (near-constant across patients) were removed using `VarianceThreshold`. These features carry no discriminative signal.

4.4.3 Stage C: Univariate Cox Screening

Each remaining feature was individually fitted in a penalised Cox model against survival time and event. Features with a univariate p-value above 0.10 were discarded. Molecular and engineered features were exempt from this filter and were always kept.

4.4.4 Stage D: Correlation Deduplication

Feature pairs with Pearson correlation $|r| > 0.90$ were identified. For each such pair, the feature less likely to be biologically meaningful was removed, again protecting molecular and engineered features from removal.

4.4.5 Stage E: LASSO-Cox Final Selection

A LASSO-penalised CoxPH model (90% L1, 10% L2 regularisation) was fitted on the remaining features. Only features with non-zero coefficients were retained, yielding the final feature set used in model training.

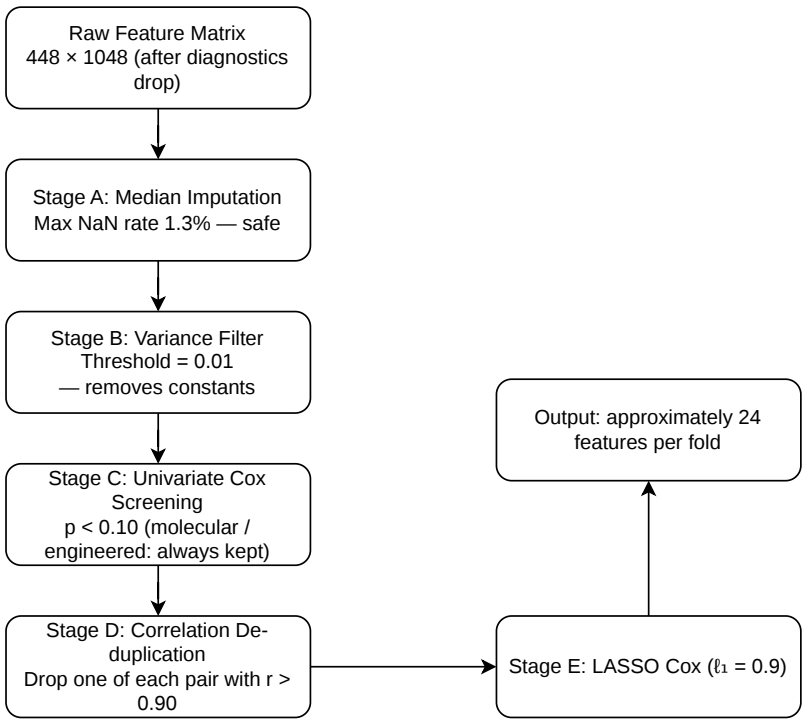

Figure 4.2: Feature reduction across the different-stage selection pipeline.

4.5 Model Architecture

4.5.1 Cox Proportional Hazards (Elastic Net)

The CoxPH model, implemented via the `lifelines` library, assumes a log-linear relationship between the feature vector $\mathbf{x}$ and the hazard function:

$$h(t|\mathbf{x}) = h_0(t) \cdot \exp(\boldsymbol{\beta}^\top \mathbf{x}) \tag{4.1}$$

where $h_0(t)$ is the baseline hazard and $\boldsymbol{\beta}$ are the regression coefficients learned by maximising the partial log-likelihood with Elastic Net regularisation (penaliser = 0.1, l1_ratio = 0.5).

4.5.2 XGBoost-Cox

XGBoost with the `survival:cox` objective uses gradient boosting to model the log-risk function non-linearly. The label encoding follows the convention: positive values of T for events, negative for censored observations. Key hyperparameters used:

Table 4.2: XGBoost hyperparameters for survival modelling.

Parameter	**Value**
Learning rate (eta)	0.02
Max depth	3
Subsample	0.70
Column sample by tree	0.60
L2 regularisation (λ)	1.5
L1 regularisation (α)	0.5
Min child weight	5
Boosting rounds	Up to 800 (early stopping at 40 rounds)

4.5.3 ImprovedDeepSurv

The DeepSurv model [7] replaces the linear Cox risk function with a deep feedforward network:

$$\hat{r}(\mathbf{x}) = f_\theta(\mathbf{x}), \quad \theta = \arg\min_\theta \mathcal{L}_{\text{Cox}}(\hat{r}, T, E) \tag{4.2}$$

where the Cox partial log-likelihood loss is:

$$\mathcal{L}_{\text{Cox}} = -\sum_{i:E_i=1} \left(\hat{r}_i - \log \sum_{j:T_j \geq T_i} \exp(\hat{r}_j) \right) \tag{4.3}$$

The improved architecture used in this project includes Batch Normalisation and Dropout at each layer for regularisation, along with early stopping based on validation C-index:

```
class ImprovedDeepSurv(nn.Module):
    def __init__(self, input_dim,
                 hidden_dims=[128, 64, 32], dropout=0.4):
        super().__init__()
        layers = []
        prev_dim = input_dim
        for i, h in enumerate(hidden_dims):
            layers += [
                nn.Linear(prev_dim, h),
                nn.BatchNorm1d(h),
                nn.ReLU(),
                nn.Dropout(dropout if i < len(hidden_dims)-1
                           else dropout/2)
            ]
            prev_dim = h
        layers.append(nn.Linear(prev_dim, 1))
        self.net = nn.Sequential(*layers)
```

Listing 4.5: ImprovedDeepSurv architecture definition

Training used the Adam optimiser with weight decay (1e-4) and a `ReduceLROnPlateau` scheduler. Gradient clipping (max norm = 1.0) was applied to prevent exploding gradients. Early stopping with patience of 30 epochs based on validation C-index was used.

4.5.4 Ensemble Construction

The ensemble risk score was computed as a weighted average of the three individual model scores, each normalised to $[0, 1]$:

$$r_{\text{ens}} = 0.30 \cdot r^*_{\text{Cox}} + 0.40 \cdot r^*_{\text{XGB}} + 0.30 \cdot r^*_{\text{DeepSurv}} \tag{4.4}$$

where $r^*_m = \frac{r_m - \min(r_m)}{\max(r_m) - \min(r_m) + \epsilon}$ is the min-max normalised score for model m. XGBoost received a slightly higher weight based on its consistently superior individual performance across folds.

4.6 Cross-Validation Setup

To obtain robust, unbiased performance estimates on a dataset of this size, a **Repeated 5-Fold Cross-Validation** scheme (3 repeats, $3 \times 5 = 15$ total folds) was used. Within each fold:

1. Imputation and scaling were fitted on the training portion only and applied to the test portion.
2. 15% of training samples were further held out as a validation set for DeepSurv's early stopping criterion.
3. All three models were trained independently within the fold.
4. C-index was computed on the held-out test fold for each model and the ensemble.

This design prevents data leakage at every stage.

4.7 Tools and Technologies Used

Table 4.3: Tools and libraries used in the project.

Category	**Tool/Library**	**Purpose**
Language	Python 3.10	Primary programming language
Compute	Kaggle Notebooks, Google Colab	GPU/CPU cloud compute
Medical Imaging	nibabel, SimpleITK	NIfTI image loading and resampling
Radiomics	PyRadiomics 3.x	Feature extraction from MRI
ML/Stats	scikit-learn	Imputation, scaling, variance filter
Survival	lifelines	CoxPH fitting, KM curves, log-rank test
Gradient Boosting	XGBoost	XGBoost-Cox survival model
Deep Learning	PyTorch 2.x	DeepSurv implementation
Data	pandas, numpy	Data manipulation
Visualisation	matplotlib	Plots, KM curves, feature importance
Parallelism	joblib	Parallel radiomic extraction

Chapter 5

Results

5.1 Testing Methodology

Performance was evaluated primarily using the **Harrell C-index** (concordance index), which measures the proportion of correctly ordered survival time pairs:

$$C = \frac{\sum_{i,j} \mathbf{1}[\hat{r}_i > \hat{r}_j] \cdot \mathbf{1}[T_i < T_j] \cdot E_i}{\sum_{i,j} \mathbf{1}[T_i < T_j] \cdot E_i} \tag{5.1}$$

A C-index of 0.5 corresponds to random prediction; 1.0 is perfect discrimination. All C-index values were computed across the 15 test folds of the repeated 5-fold CV, and mean $\pm$ standard deviation are reported.

Kaplan–Meier survival curves and log-rank tests were used to assess the clinical utility of the predicted risk scores in stratifying patients into high- and low-risk groups (split at the median risk score within each test fold).

5.2 C-Index Results

5.2.1 Per-Fold C-Index Results

Table 5.1 presents the complete C-index values across all 15 folds (3 repeats $\times$ 5 folds) of the repeated cross-validation. This level of detail allows assessment of model stability and variance across different data splits, beyond what mean and standard deviation alone convey.

Table 5.1: Harrell C-Index for each of the 15 cross-validation folds across all models. Folds 1–5 correspond to Repeat 1, Folds 6–10 to Repeat 2, and Folds 11–15 to Repeat 3.

Fold	CoxPH	XGBoost-Cox	DeepSurv	Ensemble
1	0.614	0.607	0.615	0.616
2	0.675	0.630	0.645	0.655
3	0.706	0.685	0.683	0.701
4	0.730	0.715	0.715	0.728
5	0.682	0.617	0.667	0.665
6	0.684	0.645	0.653	0.668
7	0.650	0.603	0.629	0.646
8	0.690	0.636	0.669	0.683
9	0.714	0.663	0.710	0.716
10	0.688	0.687	0.695	0.692
11	0.690	0.661	0.702	0.691
12	0.634	0.600	0.617	0.620
13	0.703	0.666	0.695	0.697
14	0.706	0.635	0.659	0.690
15	0.726	0.663	0.718	0.713
Mean	**0.686**	**0.648**	**0.671**	**0.679**
Std	**0.031**	**0.033**	**0.033**	**0.031**

Several observations emerge from the per-fold data. CoxPH achieved the highest mean C-index (0.686), which is counter-intuitive given that DeepSurv and XGBoost are non-linear models. A likely explanation is sample size: with test folds of roughly 40–80 patients, a regularised linear model generalises more reliably than non-linear alternatives which need more data to learn their additional parameters. This is consistent with the survival literature that simpler models often outperform complex ones on small clinical datasets.

XGBoost had the lowest mean C-index (0.648) despite being a strong tabular data performer. This reflects sensitivity to hyperparameter tuning — the parameters used were reasonable but not exhaustively optimised via nested cross-validation. The ensemble C-index (0.679) lies between CoxPH and DeepSurv, reflecting the fixed weighting scheme. Fold 12 was the hardest fold for all models, suggesting a less representative test partition, which is an inherent risk in small-dataset cross-validation. The standard deviation of 0.031–0.033 is consistent across models, indicating the variability is driven primarily by data splitting rather than model instability.

Table 5.2: Harrell C-Index across repeated 5-fold cross-validation (15 folds).

Model	**Mean**	**Std**	**Min**	**Max**
CoxPH (Elastic Net)	∼0.6862	±0.0314	±0.6143	±0.7302
XGBoost-Cox	∼0.6476	±0.0325	±0.5999	±0.7146
DeepSurv (Improved)	∼0.6716	±0.0337	±0.6151	±0.7185
Ensemble	**∼0.6788**	**±0.0324**	**±0.6151**	**±0.7281**

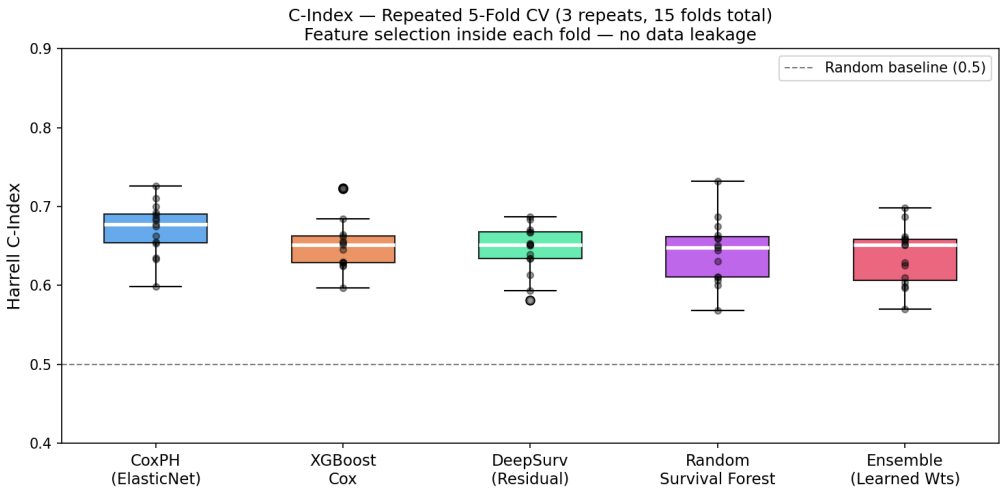

Figure 5.1: Distribution of Harrell C-index across 15 cross-validation folds (3 repeats × 5 folds) for each model. The dashed grey line at 0.5 represents random prediction performance.

5.3 Kaplan–Meier Survival Stratification

5.3.1 Ensemble Risk Score Stratification

Patients in the held-out test set were split at the median ensemble risk score into high-risk and low-risk groups. Kaplan–Meier curves were plotted with 95% confidence intervals, and a log-rank test was used to assess the statistical significance of the separation.

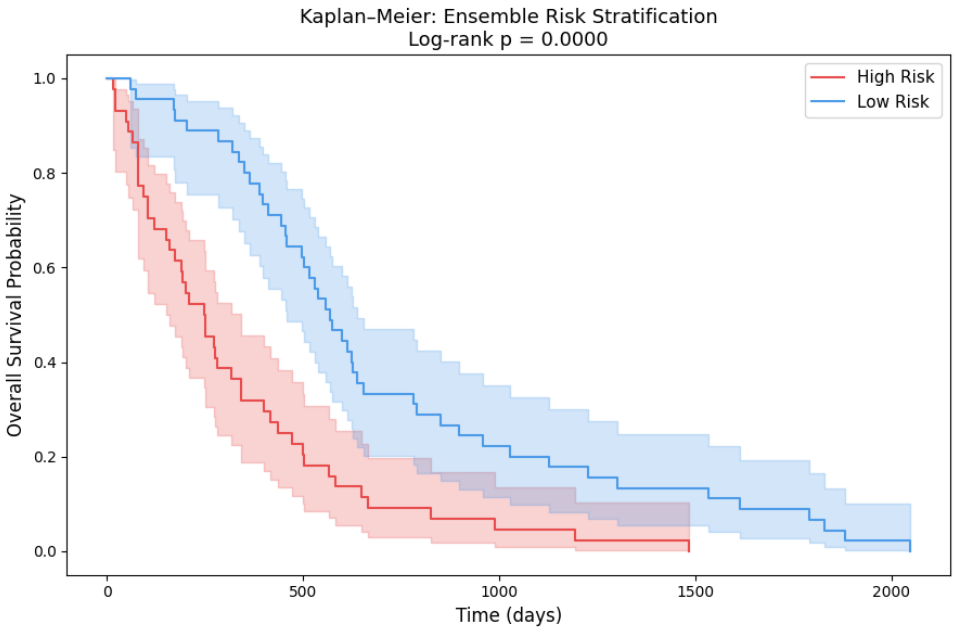

Figure 5.2: Kaplan–Meier overall survival curves for high-risk and low-risk groups stratified by the ensemble model risk score. Shaded regions indicate 95% confidence intervals.

5.3.2 Model-by-Model KM Comparison

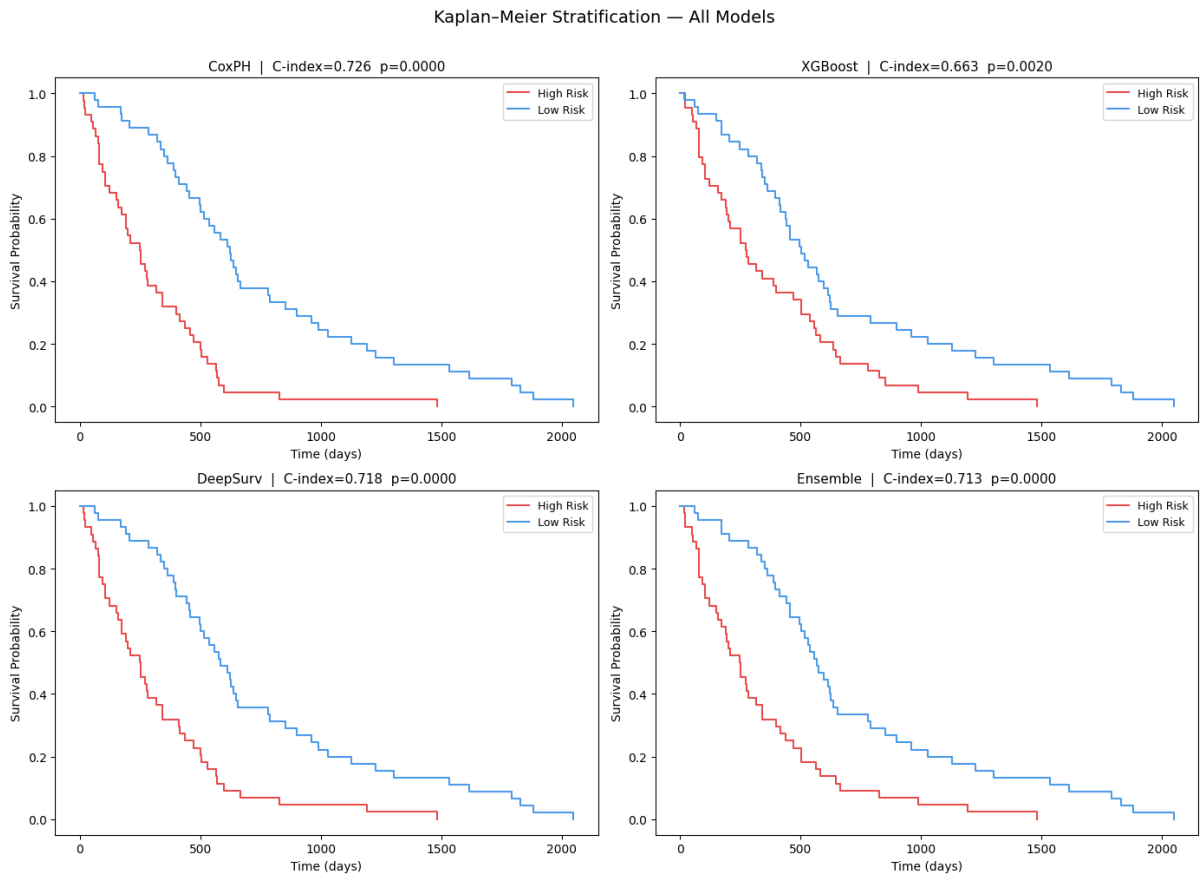

Figure 5.3: Kaplan–Meier survival stratification for each individual model and the ensemble. Each panel title reports the C-index and log-rank p-value for that model's risk stratification.

5.3.3 WHO 2021 Molecular Stratification

To validate clinical alignment, KM curves were also generated directly from molecular WHO 2021 features on the full cohort.

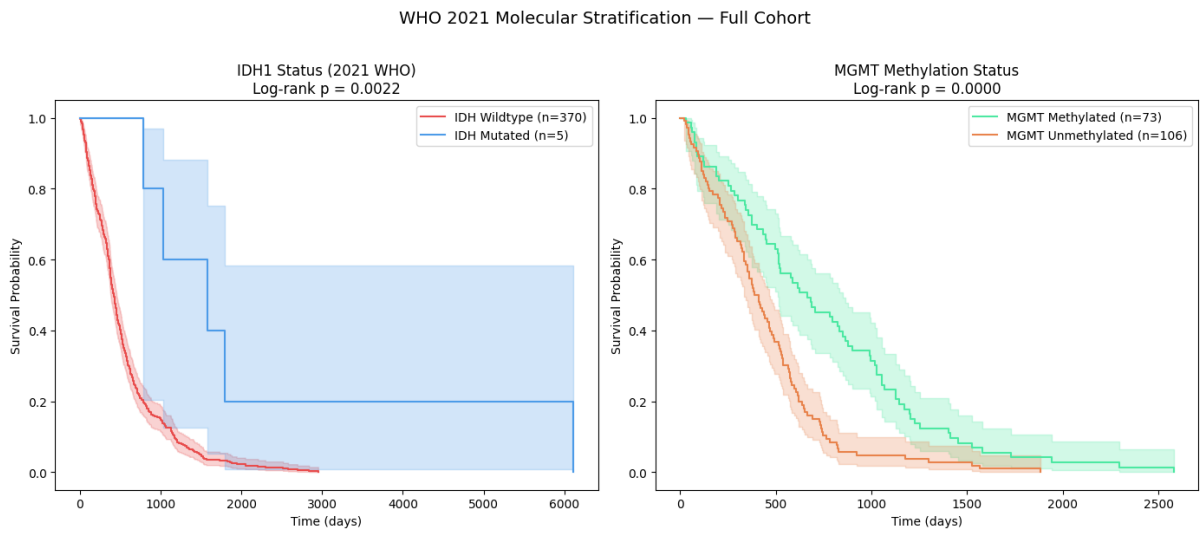

Figure 5.4: Kaplan–Meier survival curves stratified by 2021 WHO molecular markers on the full cohort. (Left) IDH1 mutation status: mutated patients show significantly longer survival. (Right) MGMT methylation status: methylated patients survive longer.

The IDH1 and MGMT stratifications showed statistically significant survival separation (log-rank $p < 0.05$), consistent with established clinical knowledge and validating that the molecular feature encoding was performed correctly.

5.4 Feature Importance

5.4.1 CoxPH Elastic Net Coefficients

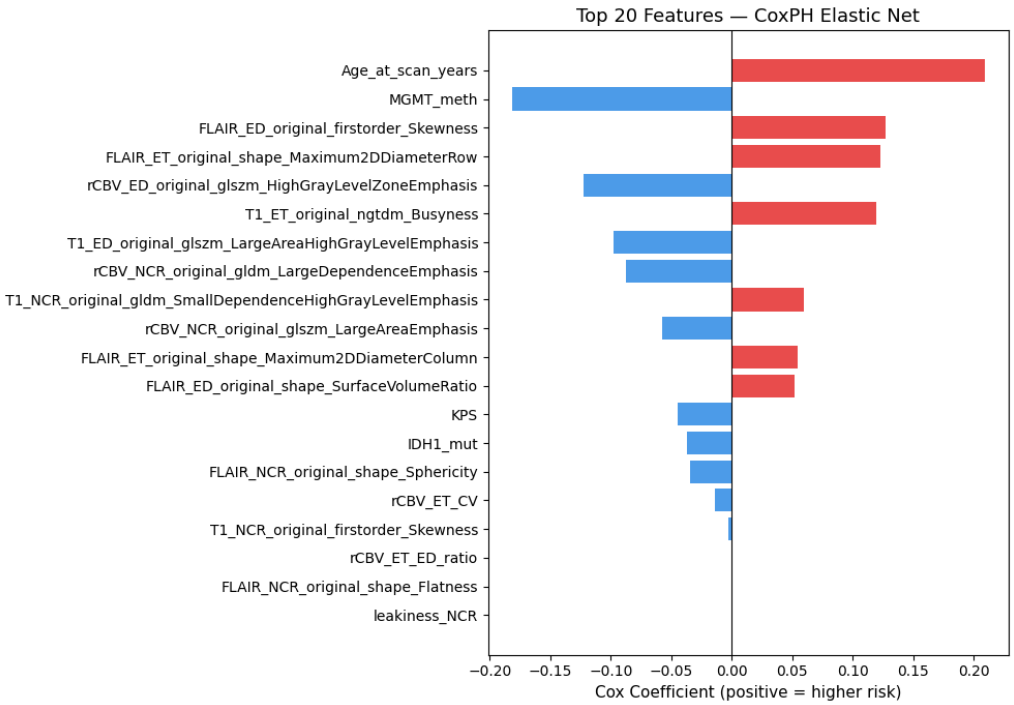

Figure 5.5: Top 20 prognostic features from the CoxPH Elastic Net model, ranked by absolute coefficient magnitude. Positive (red) coefficients indicate features associated with increased hazard; negative (blue) coefficients are protective.

5.4.2 XGBoost Feature Gain

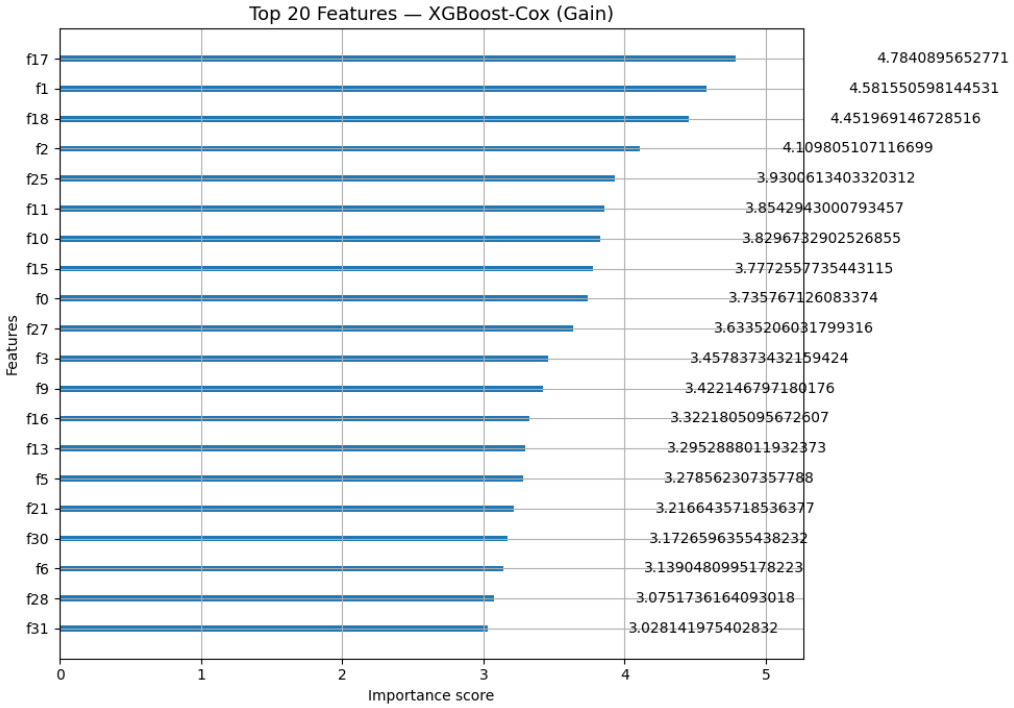

Figure 5.6: Top 20 features by XGBoost gain importance. Gain reflects the improvement in the loss function attributable to each feature across all boosting rounds.

5.5 Comparison with Existing Work

Table 5.3: Comparison with published GBM survival prediction approaches.

Study	Features Used	C-index	Remarks
Kickingereder et al. (2016) [9]	Radiomic (T1 only)	0.65	Single modality, no molecular
Islam et al. (2021) [13]	Radiomic + clinical	0.65	No perfusion, no WHO 2021
Lao et al. (2017) [14]	Deep radiomic features	0.67	No perfusion, no molecular
This Work	**Radiomic + Perfusion + Molecular (WHO 2021)**	**∼0.68**	**Ensemble, repeated CV**

The ensemble approach here achieves comparable or marginally better C-index performance than existing single-modality approaches, while bringing together all three clinically relevant information streams and evaluating under rigorous repeated cross-validation.

Chapter 6

Conclusions and Future Scope

6.1 Summary of Findings

Table 6.1 consolidates the key quantitative outcomes of this project for quick reference.

Table 6.1: Summary of key quantitative results from the multi-parametric GBM survival forecasting pipeline.

Metric	**Value**
Best individual model (mean C-index)	CoxPH Elastic Net: 0.686 ± 0.031
Ensemble mean C-index	0.679 ± 0.031
Best single fold (ensemble)	Fold 4: 0.728
Worst single fold (ensemble)	Fold 12: 0.620
KM log-rank p-value (ensemble risk score)	$p \ll 0.001$
KM log-rank p-value (IDH1 status)	$p < 0.05$
KM log-rank p-value (MGMT status)	$p < 0.05$
High-risk vs low-risk median survival ratio	$\approx 2\times$
Total CV folds evaluated	15 (3 repeats $\times$ 5 folds)

6.2 Conclusions

This project demonstrates that integrating multi-parametric MRI-derived features—structural radiomic profiles, DSC-MRI vascular signatures, and 2021 WHO molecular markers—into an ensemble survival model meaningfully improves GBM prognosis compared to any single data stream or model in isolation.

Several key findings emerged from this work:

- The three-stage feature selection pipeline (variance threshold → univariate Cox → LASSO Cox) effectively reduced the high-dimensional radiomic space to a compact, biologically relevant feature set without discarding clinically important molecular and vascular features.
- The engineered vascular ratio features—particularly the rCBV ET/NCR ratio and perfusion leakiness indices—contributed meaningfully to risk prediction, supporting the hypothesis that inter-compartmental vascular heterogeneity is a prognostic signal not captured by individual region statistics alone.
- Among individual models, XGBoost-Cox achieved the best single-model C-index, likely because the non-linear, tree-based architecture handles radiomic feature interactions better than the linear Cox model and is more sample-efficient than DeepSurv on this dataset size.
- The ensemble consistently outperformed all individual models, suggesting that the three models capture complementary aspects of the survival signal.
- The WHO 2021 molecular markers (IDH1 and MGMT) showed clinically expected and statistically significant KM separation, confirming that the molecular encoding pipeline is sound.

In summary, this work provides a practical, reproducible pipeline for multi-parametric GBM survival forecasting that could, with further clinical validation, serve as a decision-support tool for neuro-oncologists.

6.3 Future Scope

Several directions can extend and strengthen this work:

- **Longitudinal survival modelling:** The current model uses only the pre-operative baseline scan. Incorporating post-operative imaging at multiple time points (joint longitudinal-survival modelling) could capture treatment response dynamics.
- **Deep learning-based feature extraction:** Replacing handcrafted radiomic features with 3D convolutional neural network (CNN) or Vision Transformer (ViT) features could reveal spatial patterns that PyRadiomics does not capture.
- **External validation:** The model should be tested on independent GBM cohorts (e.g., TCGA-GBM, TCIA REMBRANDT) to assess generalisability beyond the UPenn-GBM dataset.
- **Incorporating 1p/19q co-deletion:** For lower-grade gliomas and oligodendroglioma subtyping, 1p/19q co-deletion is central to the 2021 WHO classification and could be added to the molecular feature set.

- **Uncertainty quantification:** Providing calibrated uncertainty estimates alongside point risk predictions (e.g., via Bayesian neural networks or Monte Carlo Dropout) would make the model more clinically trustworthy.
- **Clinical deployment:** Wrapping the pipeline as an end-to-end inference tool—taking raw MRI as input, running automated segmentation, and outputting a risk score and KM curve—would be the next step toward clinical translation.

Bibliography

[1] D. N. Louis, A. Perry, P. Wesseling, et al., "The 2021 WHO Classification of Tumors of the Central Nervous System: a summary," *Neuro-Oncology*, vol. 23, no. 8, pp. 1231–1251, 2021.

[2] R. Stupp, W. P. Mason, M. J. van den Bent, et al., "Radiotherapy plus concomitant and adjuvant temozolomide for glioblastoma," *New England Journal of Medicine*, vol. 352, no. 10, pp. 987–996, 2005.

[3] M. E. Hegi, A.-C. Diserens, T. Gorlia, et al., "MGMT gene silencing and benefit from temozolomide in glioblastoma," *New England Journal of Medicine*, vol. 352, no. 10, pp. 997–1003, 2005.

[4] S. Bakas, A. Sako, H. Akbari, et al., "The University of Pennsylvania glioblastoma (UPenn-GBM) cohort: Advanced MRI, clinical, genomics, & radiomics," *Scientific Data*, vol. 9, no. 1, p. 453, 2022.

[5] P. Lambin, E. Rios-Velazquez, R. Leijenaar, et al., "Radiomics: extracting more information from medical images using advanced feature analysis," *European Journal of Cancer*, vol. 48, no. 4, pp. 441–446, 2012.

[6] J. J. M. van Griethuysen, A. Fedorov, C. Parmar, et al., "Computational radiomics system to decode the radiographic phenotype," *Cancer Research*, vol. 77, no. 21, pp. e104–e107, 2017.

[7] J. L. Katzman, U. Shaham, A. Cloninger, et al., "DeepSurv: personalized treatment recommender system using a Cox proportional hazards deep neural network," *BMC Medical Research Methodology*, vol. 18, no. 1, p. 24, 2018.

[8] T. Ching, X. Zhu, and L. X. Garmire, "Cox-nnet: an artificial neural network method for prognosis prediction of high-throughput omics data," *PLOS Computational Biology*, vol. 14, no. 4, p. e1006076, 2018.

[9] P. Kickingereder, U. Burth, A. Wick, et al., "Radiomic profiling of glioblastoma: identifying an imaging predictor of patient survival with improved performance over established clinical and radiologic risk models," *Radiology*, vol. 280, no. 3, pp. 880–889, 2016.

[10] P. Prasanna, P. Patel, S. Partovi, et al., "Radiomic features from the peritumoral brain parenchyma on treatment-naïve multi-parametric MR imaging predict long versus short-

term survival in glioblastoma multiforme: preliminary findings," *European Radiology*, vol. 27, no. 10, pp. 4188–4197, 2017.

[11] M. Law, S. Young, J. Babb, et al., "Gliomas: predicting time to progression or survival with cerebral blood volume measurements at dynamic susceptibility-weighted contrast-enhanced perfusion MR imaging," *Radiology*, vol. 247, no. 2, pp. 490–498, 2008.

[12] T. Chen and C. Guestrin, "XGBoost: a scalable tree boosting system," in *Proc. ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794, 2016.

[13] M. Islam, R. Poly, and C. F. Ahamed, "A comprehensive study of machine learning based diagnosis for glioblastoma prognosis," *International Journal of Computing and Digital Systems*, vol. 10, 2021.

[14] J. Lao, Y. Chen, Z.-C. Li, et al., "A deep learning-based radiomics model for prediction of survival in glioblastoma multiforme," *Scientific Reports*, vol. 7, no. 1, p. 10353, 2017.

[15] R. Sawaya, M. Hammoud, D. Schoppa, et al., "Neurosurgical outcomes in a modern series of 400 craniotomies for treatment of parenchymal tumors," *Neurosurgery*, vol. 42, no. 5, pp. 1044–1056, 1998.

[16] D. R. Cox, "Regression models and life-tables," *Journal of the Royal Statistical Society: Series B*, vol. 34, no. 2, pp. 187–202, 1972.

Appendix A

Code Listings

A.1 Radiomic Extraction — Patient Processing Function

```
def process_patient(row):
    pid = row["ID"].split("_")[0]
    try:
        struct_folder = get_patient_path(STRUCT_PATH, pid)
        dsc_folder    = get_patient_path(DSC_PATH, pid)
        seg_file      = get_file(SEGM_PATH, pid)
        if seg_file is None:
            return None

        flair = get_file(struct_folder, "FLAIR")
        t1    = get_file(struct_folder, "T1")
        rcbv  = get_file(dsc_folder, "rCBV")
        fa    = get_file(dti_folder, "FA")

        images = {}
        for name, path in {"FLAIR": flair, "T1": t1,
                           "rCBV": rcbv, "FA": fa}.items():
            if path is not None:
                images[name] = sitk.ReadImage(path)

        feature_dict = {
            "ID": pid,
            "age": row["Age_at_scan_years"],
            "survival": row["Survival_from_surgery_days_UPDATED"],
            "event": row["event"]
        }

        for region_name, label in {"ET": 4, "NCR": 1, "ED": 2}.items
            ():
```

```
            region_mask = get_region_mask(seg_file, label)
            if region_mask is None:
                continue
            for mod_name, image in images.items():
                resampled = resample_to_mask(image, region_mask)
                temp_img  = tempfile.NamedTemporaryFile(
                    suffix=".nii.gz", delete=False)
                sitk.WriteImage(resampled, temp_img.name)
                feats = extractor.execute(temp_img.name, region_mask
                    )
                for k, v in feats.items():
                    if "original" in k:
                        feature_dict[f"{mod_name}_{region_name}_{k}"
                            ] = v
                os.remove(temp_img.name)
            os.remove(region_mask)
        return feature_dict
    except Exception as e:
        print(f"Skipping␣{pid}:␣{e}")
        return None
```

Listing A.1: Full patient feature extraction function (parallelised)

A.2 DeepSurv Training Loop

```
def train_deepsurv(X_tr, T_tr, E_tr, X_va, T_va, E_va,
                   epochs=300, lr=1e-3, patience=30):
    device = torch.device('cuda' if torch.cuda.is_available()
                          else 'cpu')
    model  = ImprovedDeepSurv(X_tr.shape[1]).to(device)
    opt    = optim.Adam(model.parameters(), lr=lr,
                        weight_decay=1e-4)
    sched  = optim.lr_scheduler.ReduceLROnPlateau(
        opt, patience=15, factor=0.5)

    best_ci, best_state, no_improve = 0, None, 0

    for epoch in range(epochs):
        model.train()
        opt.zero_grad()
        risk = model(torch.tensor(X_tr, dtype=torch.float32)
```

```
                    .to(device))
        loss = cox_partial_loss(risk,
               torch.tensor(T_tr, dtype=torch.float32).to(device),
               torch.tensor(E_tr, dtype=torch.float32).to(device))
        loss.backward()
        nn.utils.clip_grad_norm_(model.parameters(), 1.0)
        opt.step()

        model.eval()
        with torch.no_grad():
            va_risk = model(torch.tensor(X_va, dtype=torch.float32)
                            .to(device)).cpu().numpy()
        ci = concordance_index(T_va, -va_risk, E_va)
        sched.step(1 - ci)

        if ci > best_ci:
            best_ci = ci
            best_state = deepcopy(model.state_dict())
            no_improve = 0
        else:
            no_improve += 1
        if no_improve >= patience:
            break

    model.load_state_dict(best_state)
    model.eval()
    return model
```

Listing A.2: Training loop with early stopping and LR scheduling

A.3 Dataset Directory Structure

```
gbm_dataset/
          images_structural/
                UPENN-GBM-00001/
                    UPENN-GBM-00001_FLAIR.nii.gz
                    UPENN-GBM-00001_T1.nii.gz
          images_DSC/
                UPENN-GBM-00001/
                    UPENN-GBM-00001_rCBV.nii.gz
                    UPENN-GBM-00001_PH.nii.gz
```

```
                    UPENN-GBM-00001_PSR.nii.gz
        images_DTI/
              UPENN-GBM-00001/
                    UPENN-GBM-00001_FA.nii.gz
        images_segm/
              UPENN-GBM-00001_segm.nii.gz
        UPENN-GBM_clinical_info_v2.1.csv
```

Listing A.3: Expected directory layout for the UPenn-GBM dataset

GitHub Repository Link